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(see 6.3)

MILITARY SPECIFICATION

CANDY AND CHOCOLATE CONFECTIONS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers candy and chocolate confections for use by the Armed Forces as components of operational rations and as items of general issue.

1.2 Classification. The candy shall be of the following types, styles, classes, and flavors, as specified (see 6.1).

Type I - Chocolate coated candy

Class 1 - Coconut
Class 2 - Chocolate fudge
Class 3 - Vanilla fudge

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8925

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Type II - Starch jelly candies

Flavor 1 - Assorted (orange, lemon, and lime for bars)
Flavor 2 - Assorted (orange, lemon, lime, pineapple, and cherry for bulk packaging)

Type III - Caramel and toffee

Flavor 1 - Chocolate (for rolls)
Flavor 2 - Assorted (vanilla, chocolate, rum butter, and coconut for bulk packaging)
Flavor 3 - Vanilla

Type IV - Lozenges, peppermint

Type V - Hard candy

Class 1 - Fruit tablets
Class 2 - Plastic filled, pulled jacket
Class 3 - Mint tablets

Flavor 1 - Assorted (see 3.3.1.5.3 for class 1 product)
Flavor 2 - Assorted (see 3.3.1.5.4 for class 2 product)

Type VI - Sugar, pan-coated confections

Class 1 - Peanuts
Class 2 - Jelly beans
Class 3 - Chocolate disks
Class 4 - Licorice lozenges
Class 5 - Creams
Class 6 - Almond
Class 7 - Bridge mix

Type VII - Enriched sweet chocolate, with almonds, buttercrunch, or toffee

Type VIII - Fudge bar, chocolate, survival

Styles

1 - Bars (applicable to types I, II, III, IV, V, VII, and VIII)
2 - Rolls (applicable to type III)
3 - Disks (applicable to types I and VII)
4 - Pieces (applicable to types III, IV, V, and VI)
5 - Bags (applicable to type VI)

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2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-C-110	- Cellophane (Coated and Noncoated Regenerated Cellulose Film)
L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin
QQ-A-1876	- Aluminum Foil
TT-C-495	- Coatings, Exterior for Tinned Food Cans
PPP-B-566	- Boxes, Folding, Paperboard
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-B-676	- Boxes, Setup
PPP-C-96	- Cans, Metal, 28 Gage and Lighter

MILITARY

MIL-L-1497	- Labeling of Metal Cans for Subsistence Items
MIL-L-10547	- Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible
MIL-L-35078	- Loads, Unit: Preparation of Semiperishable Subsistence Items, Clothing Personal Equipment and Equipage; General Specification For

STANDARDS

FEDERAL

FED-STD-101	- Test Procedures for Packaging Materials

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage

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(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

OTHER GOVERNMENT DOCUMENTS

U.S. DEPARTMENT OF AGRICULTURE (USDA)

U.S. Standards for Grades of Shelled Spanish Type Peanuts

(Application for copies should be addressed to the U.S. Department of Agriculture, Agricultural Marketing Service, Fruit and Vegetable Division, Fresh Products Branch, Room 2056, South Building, Washington, DC 20250.)

Meat and Poultry Inspection Regulations

Regulations Governing the Inspection of Eggs and Egg Products (7 CFR Part 59)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. Standards for Dry Whey

U.S. Standards for Grades of Butter

U.S. Standards for Grades of Nonfat Dry Milk (Spray Process)

(Application for copies should be addressed to U.S. Department of Agriculture, Agricultural Marketing Service, Dairy Standardization Section, Dairy Division, Room 2750 South, Washington, DC 20250.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Definitions and Standards of Identity of the Food and Drug Administration (FDA)

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

FDA Bacteriological Analytical Manual

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

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2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

AMERICAN OIL CHEMISTS' SOCIETY (AOCS)

Official and Tentative Methods of the American Oil Chemists

(Application for copies should be addressed to the American Oil Chemists' Society, 508 South Sixth Street, Champaign, IL 61820.)

ASSOCIATION OF VITAMIN CHEMISTS

Methods of Vitamin Assay

(Application for copies should be addressed to the Interscience Publishers, Division of John Wiley and Sons, Inc., 605 Third Avenue, New York, NY 10016.)

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP) and the National Formulary (NF)

(Application for copies should be addressed to the United States Pharmacopeial Convention, Inc., 12601 Twinbrook Parkway, Rockville, MD 20852.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue NW, Washington, DC 20418.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 87 - Melting Point of Petroleum Wax (Cooling Curve)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Bid sample approval. Bid sample requirements shall be for type I, flavor assortment 2 of type II, type III, type IV, type V, type VI, type VII, and type VIII. Four samples of at least 4 pounds of each type, style, class, and flavor (as applicable), representative of the products which the bidder proposes to furnish, shall be submitted to the contracting officer prior to bid opening and will be tested to the extent necessary to properly evaluate the bid. Testing of bid samples will include panel testing for palatability (see 6.4). Each bid sample shall be labeled with type, style, class, and flavor, as applicable, as well as weight. For types I and VII, those candies that are found acceptable for palatability shall be saved and frozen at -20°F in suitable containers and chosen at random for use as standards in determining compliance of future bid samples, as to palatability acceptance. The standard shall be thawed at room temperature at least 24 hours before testing. Testing and acceptance of bid samples will include panel testing for palatability as specified (see 6.4), except for type II, flavor 2; type III, flavor 2; type V, classes 2 and 3; and type VI, class 7, which will be evaluated by a Government technical panel to determine acceptability. The approved bid sample shall be used as a standard for palatability of the delivered product. Approval of any subject candies for palatability does not constitute its approval as meeting other requirements of this document.

3.2 Ingredients. All the ingredients shall be clean, sound, wholesome, and free from any foreign material, evidence of rodent and insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Acids. Citric or malic acid may be used interchangeably to achieve desired flavor except as noted and shall comply with the Food Chemicals Codex. When anhydrous citric acid is used, it shall be added at a level of 91.4 percent of the hydrous quantity specified.

3.2.2 Almonds. Almonds shall be well-dried, fumigated, and properly cleaned, and shall be free from rancidity, mold, bird or insect damage, decay, worm injury, and damage caused by shriveling. The cold pressed oil of diced

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almonds prior to roasting shall contain not more than 0.3 percent free fatty acid, expressed as oleic acid (see 4.4.1). For type VI, class 6, only "Ne plus" or "Jordanola-type" almonds, of an average count not exceeding 26 per ounce, shall be used. For type VII, diced almond pieces shall be used. The pieces shall be of such size that not more than 1 percent shall be retained on a 16/64-inch round-hole screen or a 12/64-inch round-hole screen, or of such size that not more than 2 percent shall pass through a 6/64-inch or 1 percent through a 4/64-inch size screen.

3.2.3 Antioxidant. Antioxidant mixture shall consist of ingredients in the proportions specified in table I. Unless otherwise specified, any one of the mixtures may be used. Antioxidants shall comply with the Food Chemicals Codex.

TABLE I. Antioxidant mixture

Ingredient	Percent of mixture				
	A	B	C	D	E
Edible solvent, not more than	70	74	60	67	70
Butylated hydroxyanisole, not less than	20	20	20	4	20
Butylated hydroxytoluene, not less than	--	--	20	20	--
Citric acid	4	6	--	4	4
Propyl gallate, not less than	6	--	--	5	--
TBHQ (Tertiary Butylhydroquinone) not less than	--	--	--	--	6

3.2.4 Butter. Butter at time of use shall be U.S. Grade A or better, as defined in U.S. Standards for Grades of Butter.

3.2.5 Chocolate liquor. Chocolate liquor shall conform to the standards of identity of the Food and Drug Administration. It shall be manufactured from roasted cocoa beans equal to or better in quality and flavor than fair, fermented Ghana; fair, average quality Lagos; or superior Bahia.

3.2.6 Sweet chocolate and sweet milk chocolate. Sweet chocolate and sweet milk chocolate shall meet the requirements of standardized chocolate in CFR, Part 163 - Cocoa Powder (see 3.3.1.6.3).

3.2.7 Cocoa butter. Cocoa butter shall be that obtained by hydraulic pressure from roasted sound, cocoa beans.

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3.2.8 Cocoa powder. The cocoa powder shall conform to the standards of identity for medium fat cocoa of the Food and Drug Administration. Not less than 98 percent by weight, shall pass through a U.S. Standard No. 200 sieve.

3.2.9 Coconut. Coconut shall be of the best quality macaroon or shred type that has been tenderized or sweetened. It shall be white, clean, and shall contain no foreign material such as specks or dirt. It shall have characteristic coconut odor and taste, with no musty, stale, moldy odor, or moldy taste. The sweetened tenderized coconut, upon analysis, shall contain not more than 11.5 percent moisture and not more than 0.10 percent free fatty acid (calculated as oleic acid, on a moisture-free basis). The pH shall be 6.0 or higher. Coconut shall be tenderized (moistened) before incorporation into the formula. It may be purchased as such or processed by the candy contractor.

3.2.10 Colors. Coloring materials shall be U.S. certified food colors.

3.2.11 Condensed skim milk. Condensed skim milk shall comply with the standards of identity for sweetened condensed skim milk.

3.2.12 Confectioners' glaze. Confectioners' glaze shall be of the grade and quality customarily used in the production of high quality pan-coated confections.

3.2.13 Corn syrup. Corn syrup, on the dry basis, shall contain not less than 40 percent, by weight, of reducing sugars, calculated as anhydrous dextrose. Dried corn syrup may be used. For type V hard candy, either a regular conversion high maltose 42 dextrose equivalent corn syrup or a low conversion 35-38 dextrose equivalent range corn syrup shall be used, whichever is required to produce the product set forth herein.

3.2.14 Cream of tartar. Cream of tartar shall comply with the Food Chemicals Codex.

3.2.15 Cream. Cream shall comply with the standards of identity for light cream.

3.2.16 Dextrose. Dextrose shall comply with the Food Chemicals Codex.

3.2.17 Egg whites, dried. The dried egg whites shall be processed in accordance with 7 CFR part 59, Regulations Governing the Inspection of Eggs and Egg Products, and identified with the USDA inspection shield mark. The dried egg whites shall be powdered and free flowing. It shall be salmonella negative as evidenced by USDA certification. It shall be free from objectionable odors, flavors, and/or foreign material.

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3.2.18 Emulsifiers. Lecithin, monoglycerides, or diglycerides of fatty acids, or a combination of these, may be used. These products shall comply with the Food Chemicals Codex.

3.2.19 Fat. Vegetable fat (for type I chocolate coatings for enrobing and type VII chocolate candy) shall be natural or hydrogenated coconut, palm kernel, babasu, tucum, or other high lauric acid oils or mixtures thereof, or a mixture of one or more of these with not more than 25 percent hydrogenated peanut oil or cottonseed oil, or both combined. In type VIII (fudge) and type I, classes 2 and 3 (chocolate fudge and vanilla fudge centers only), the added fat shall be only a domestic hydrogenated vegetable fat. Both the domestic hydrogenated vegetable fat and the lauric acid type fats (coconut) shall have a minimum fat stability of 100 hours when measured by the active oxygen method (AOM) (see 4.4.1). They shall retain satisfactory odor, flavor, and color after heating to a temperature of 400°F. The free fatty acid content shall be not greater than 0.08 percent (prior to the addition of antioxidant mixture) or 0.11 percent after the addition of antioxidant mixture A, B, C, D, or E (see 3.2.3, table I). The moisture and volatile matter shall not exceed 0.1 percent after the addition of antioxidant mixture A, B, C, D, or E (see 4.4.1). The fat shall be adequately protected against oxidative rancidity, at time of manufacture or by the candy processor, by the addition of 0.10 percent by weight of an antioxidant mixture specified in 3.2.3. The melting point shall be in accordance with the formulation requirements.

3.2.20 Flavors. Flavors shall comply with the following:

a. Orange shall be domestic cold-pressed orange oil and shall comply with the Food Chemicals Codex.
b. Lemon shall be domestic cold-pressed lemon oil and shall comply with the Food Chemicals Codex.
c. Lime shall be distilled oil of lime and shall comply with the Food Chemicals Codex.
d. Peppermint shall be triple distilled oil of peppermint and shall comply with the Food Chemicals Codex.
e. Wintergreen shall be pharmacopeial methyl salicylate and shall comply with the Food Chemicals Codex.
f. Cherry shall be imitation, natural, or combination of both.
g. Pineapple shall be imitation, natural, or combination of both.

Any other essential oils shall meet pharmacopeial or Food Chemicals Codex requirements. All essential oils, prior to use, shall be kept in containers adequately protected from oxidation and development of rancidity. Vanilla-type flavors shall be vanilla extract or imitation vanilla, or both combined. Imitation flavors shall be of such quality that the imitated flavor can be readily identified when the finished candy is tested for palatability (see 6.4).

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3.2.21 Gelatin. Gelatin shall be domestic and of food grade.

3.2.22 Glycerin. Glycerin shall comply with the Food Chemicals Codex.

3.2.23 Gums, natural. Gum arabic (acacia), gum tragacanth, and any other natural gums used shall be of food grade.

3.2.24 Milk, evaporated. The evaporated milk shall possess a sweet pleasing, and desirable flavor with not more than a definite cooked flavor. It shall be free from scorched, oxidized, or other objectionable tastes and odors. The evaporated milk shall be stabilized and of medium viscosity with a uniform consistency and appearance. It shall be smooth and free from fat or protein separation, lumps, clots, gel formation, coarse milk solids precipitate or sedimentation, and extraneous material. It shall possess a uniform natural white to light-cream color. The product shall comply with the composition requirements specified in 21 CFR 131.130.

3.2.25 Milk, nonfat,dry. Nonfat dry milk shall be U.S. Extra Grade as defined in the U.S. Standards for Grades of Nonfat Dry Milk (Spray Process).

3.2.26 Milk, sweetened, condensed. Sweetened condensed milk shall conform to the standards of identity for sweetened condensed milk.

3.2.27 Peanuts. Peanuts shall be Spanish type, U.S. No. 1 grade peanuts or dry blanched Jumbo Runner peanuts from the latest available crop, as determined in USDA and FDA Standards and Regulations for that type. After sorting, they shall be free from all shells or foreign material. Prior to roasting, raw peanuts shall be properly cleaned and shall be fumigated by a process giving adequate protection against infestation. Until ready for use, the fumigated peanuts shall be stored in a room or container adequately protected against infestation. Peanuts that are dry roasted prior to use shall be adequately protected against infestation until used. The peanuts shall be certified as aflatoxin negative by the USDA.

3.2.28 Polyoxyethylene 20 sorbitan monostearate. Polyoxyethylene 20 sorbitan monostearate shall comply with the Food Chemicals Codex.

3.2.29 Salt. Salt shall be non-iodized white, refined sodium chloride with or without anticaking agents. The salt used in the centers and coating for type I (fudge), classes 1, 2, and 3, and type VIII shall not contain more than the following content of trace metals: 0.5 parts per million (ppm) of copper and 1.0 ppm of iron.

3.2.30 Sorbitan monostearate. Sorbitan monostearate shall comply with the Food Chemicals Codex.

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3.2.31 Sorbitol. The sorbitol used shall comply with the Food Chemicals Codex. It shall be a 70 ± 1.0 percent aqueous solution of sorbitol and water-white in color. It shall have a sweet, pleasant taste and shall be odorless.

3.2.32 Soya whipping agents. Soya whipping agents shall be made from food grade soya grits.

3.2.33 Starch. Any type of edible starch that will give the desired results will be acceptable.

3.2.34 Stearic acid. Stearic acid shall comply with the Food Chemicals Codex.

3.2.35 Sugar. Unless otherwise specified, the term "sugar" shall apply only to granulated or powdered cane or beet sugar, liquid sugar (noninvert type), or any combination of these. The ash content of the liquid sugar (noninvert type) shall not exceed 0.15 percent of its total solids content. The ash content of granulated or powdered cane or beet sugar shall not exceed 0.04 percent of its total solids content (see 4.4.1).

3.2.36 Sugar, invert. Invert sugar shall be the product made by inverting partly refined or refined sugar. The ash content shall be not more than 0.3 percent of its total solids content (see 4.4.1).

3.2.37 Vitamins. Vitamin A shall be the refined concentrate of vitamin A ester (palmitate). When added to the chocolate or confections, it shall not impart a fishy or objectionable odor or flavor to the finished product. Ascorbic acid (vitamin C), thiamine mononitrate, and pyridoxine hydrochloride shall comply with the Food Chemicals Codex.

3.2.38 Wax. Only beeswax or carnauba wax, or both, shall be used and shall comply with the Food Chemicals Codex.

3.2.39 Propylene glycol. Propylene glycol shall comply with the Food Chemicals Codex.

3.2.40 Buttercrunch. Buttercrunch shall contain sugar (white or brown or both), butter, almonds, salt, lecithin, BHA, and BHT. A maximum of 5 percent solids shall remain when passed through a U.S. Standard No. 4 sieve, and a maximum of 20 percent solids shall pass through U.S. Standard No. 10 sieve.

3.2.41 Sugar fondant. Sugar fondant shall consist of tiny crystals of pure cane sugar that have been bonded with invert sugar to produce white, free-flowing porous particles that are resistant to caking. The sugar fondant shall contain not more than 0.04 percent ash based on solids.

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3.2.42 Vanillin Vanillin shall comply with the Food Chemicals Codex.

3.2.43 Whey. Whey shall be U.S. Extra Grade as defined in the U.S. Standards for Dry Whey.

3.2.44 Toffee. Toffee shall contain sugar, margarine, corn syrup, brown sugar, almonds, butter, artifical flavor, salt, and lecithin. A maximum of 5 percent solids shall remain when passed through a U.S. Standard No. 4 sieve, and a maximum of 20 percent solids shall pass through a U.S. Standard No. 10 sieve.

3.3 Finished product requirements. Finished product shall consist of confections of a quality equal to the document requirements for flavor, texture, size, shape, and weight. The products shall be free from objectionable tastes, odors, foreign material, and insect infestation. The paper, adhesives, and other material used for wrapping or packaging the finished material shall impart no odor or flavor to the product. The palatability shall be equal to or better than the bid sample, as applicable (see 3.1 and 6.4).

3.3.1 Formulas. Unless otherwise specified, the formulas in 3.3.1.1 through 3.3.1.8 are blended on a dry-solids basis. Evaporated skim milk, sweetened condensed skim milk, or other types of skim milk may be substituted in proper proportions where nonfat milk solids are specified.

3.3.1.1 Type I, chocolate type coated candies. The chocolate for coating shall contain ingredients in accordance with the following formula:

Ingredients	Percent by weight
Cocoa powder (10 to 14 percent cocoa fat)	Not less than 8.0
Nonfat milk solids	Not less than 12.0
Added fat (113° ±2°F, Wiley melting point (m.p.))	Not less than 30.0
Lecithin	Not more than 0.2
Sorbitan monostearate	Not more than 0.5
Polyoxyethylene 20 sorbitan monostearate	Not more than 0.5
Sugar	Not more than 48.5
Salt (per 100 pounds of coating) (see 3.2.29)	2 ounces
Vanillin (per 100 pounds of coating)	1 ounce

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Sorbitan monostearate and polyoxyethylene 20 sorbitan monostearate shall be melted and mixed with the added fat at a temperature of not less than 150°F and the mixture held at that temperature or higher for a period of at least 30 minutes. The coating shall be refined so that it has a smooth mouth feel without grittiness. The candies shall be covered with a uniform chocolate-type coating forming not less than 28 percent by weight of the finished product. The coating shall be free from irregularities, that is, cracks, chips, or rough spots. Candy showing uncoated portions of centers will not be acceptable. The coating shall be such that, when the wrapped coated bars (at least 72 hours after manufacture) are held at a temperature of 100°F for 2 hours, the wrapper can be easily removed from the bar or disk without adhering to the candy. The product shall be allowed to cool at a temperature between 40° and 70°F, for resolidification to occur, for approximately 1 hour prior to performing the test. All type I products shall test salmonella negative.

3.3.1.1.1 Class 1, coconut centers. The uncoated centers of class 1 confections shall conform to the following formula:

Ingredients	Percent by weight
Sugar	Not less than 48
Invert sugar solids	Not less than 8
Corn syrup solids	Not less than 17
Coconut 1/	Not less than 20
Sorbitol solids	Not less than 4
Propylene glycol	Not less than 2
Salt (see 3.2.29)	Not more than 0.6
Other edible ingredients	Not more than 0.4
Antioxidant mixture	Not less than 0.01

1/ Desiccated coconut shall be tenderized prior to incorporation into the candy formula (see 3.2.9).

NOTE: The moisture content of the coconut cream center of the coated bar or disk shall be not less than 7 percent nor more than 11 percent. After the coconut has been added and mixed with the other ingredients, the mixture shall be maintained at 165°F for 15 minutes or at 170°F or higher for 8 minutes.

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3.3.1.1.2 Class 2 and 3 centers. Class 2 and 3 centers shall be chocolate and vanilla fudge, respectively, and shall conform to the following formulas:

Ingredients		Percent by weight Chocolate	Vanilla
Sugar	Not less than	56	56
Corn syrup solids	Not more than	13	15
Sorbitol solids	Not less than	10	10
Fat (110° ±2°F, Wiley m.p.)	Not less than	8	11
Nonfat milk solids	Not less than	5	8
Egg whites or soya whip or combination of both	Not less than	0.1	0.1
Chocolate liquor	Not less than	8.0	-
Salt (see 3.2.29)	Not more than	0.6	0.6
Antioxidant mixture	Not less than	0.01	0.01
Emulsifiers	Not less than	0.1	0.1
Vanilla-type flavor		As needed	As needed
Imitation butter flavor		-	As needed

NOTE: The moisture content of the finished vanilla and chocolate fudge centers shall be not less than 8 percent nor more than 11 percent.

3.3.1.2 Type II, starch jelly candies. Type II candies shall be of uniform shape and size. For flavor and assortment, see 3.3.1.2.1. They shall be prepared in accordance with the following formula:

Ingredients	Percent by weight (before sanding)
Sugar	Not less than 31.0 nor more than 38.0
Corn-syrup solids	Not more than 56.0
Starch	Not more than 13.0
Other edible ingredients	Not more than 0.4
Mono- and diglycerides	Not more than 0.1

The candies shall be sanded with commercial-type sanding sugar. No acids shall be used; however, sodium citrate may be used up to 0.1 percent to adjust pH. Cream of tartar may be used as a doctor if the open cook method of cooking is practiced. The bars shall not sweat when exposed to a temperature of 100°F for 24 hours in their prescribed wrappers. The moisture content of the finished candy, after sanding, shall be not less than 14 nor more than 19 percent. The pH shall be not less than 5.0 and not more than 6.5.

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3.3.1.2.1 Assortment. Four-piece packages shall consist of two lemon, one orange, and one lime flavor. Pieces for bulk packages shall consist of approximately equal amounts of each flavor, and the count per pound shall be not less than 45 nor more than 65. Not more than 2 percent by weight of the bulk packaged product shall be broken or crushed. The colors and flavors shall conform to the composition in table II.

TABLE II. Assortment composition for type II

Color	Flavor	Flavoring to be added
Orange	Orange	Orange oil (as needed)
Yellow	Lemon	Lemon oil (as needed)
Green	Lime	Lime oil (as needed)
Red	Cherry	Imitation wild cherry (as needed)
Yellow	Pineapple	Pineapple flavor (as needed)

3.3.1.3 Type III, caramels and toffee.

3.3.1.3.1 Rolls, caramel. The type III, flavor 1 caramel rolls shall be of chocolate flavor only and shall be made in accordance with the following formula:

Ingredients	Percent by weight
Sugar	Not less than 54
Corn syrup solids	Not more than 30
Nonfat milk solids	Not less than 3.5
Fat	Not less than 6.75
Salt	Not more than 0.5
Chocolate liquor	Not less than 1.65
Cocoa powder	Not less than 3.35
Emulsifiers	Not less than 0.08
Antioxidant mixture	Not less than 0.01
Vanilla-type flavor	As needed

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NOTE: The moisture content of the finished roll shall be not less than 5 percent nor more than 8 percent. The rolls shall be scored slightly to form five distinct sections. The inner portion of the roll shall be grainy in appearance but chewy in character. The outer portion of the roll shall be ungrained and chewy.

3.3.1.3.2 Packaged toffee pieces. The type III, flavor 2 packaged assorted pieces shall consist of approximately equal amounts of vanilla, chocolate, rum butter, and coconut. The product shall not adhere to the wrappers when subjected to 100°F for 2 hours. The moisture content of the finished product shall be not less than 5 percent nor more than 8 percent. The finished product shall be uniform in shape and shall be of such size that not less than 55 nor more than 90 pieces are contained in 1 pound.

3.3.1.3.3 Caramels, vanilla-flavored. Caramels, vanilla-flavored shall contain condensed skim milk, corn syrup, sugar, hydrogenated vegetable oil, cream, whey solids, salt, and artificial and natural flavors. The product shall not adhere to wrappers when subjected to 100°F for 2 hours. The moisture content of the finished product shall be not less than 5 percent nor more than 9.5 percent.

3.3.1.4 Type IV, lozenges. Type IV lozenges shall be white in color. The moisture content shall not exceed 1.5 percent by weight of the finished product. Type IV confections shall conform to the following formula:

Ingredient	Percent by weight
Sugar	Not less than 90
Dextrose	Not more than 3.5
Corn syrup solids	Not more than 4.0
Glycerine	Not more than 1.1
Gelatin	Not more than 0.3
Natural gums	Not more than 0.2
Peppermint oil (triple distilled)	Not more than 0.2
Other ingredients	Not more than 0.7

3.3.1.5 Type V, hard candy, classes 1, 2 and 3. Classes 1, 2 (except for blend and jacket requirements), and 3 of hard candy shall be made from a blend of sugar, corn syrup, and natural and artificial flavors. Cream of tartar and citric acid are optional ingredients. For colored candies, U.S. certified colors shall be used. The assorted candy tablets of class 1, style 1, consisting of ten tablets, shall include no fewer than five of the following assortment: lemon, lime, cherry, grape, orange, and pineapple. The product shall contain not more than 2 percent moisture nor more than 19.5 percent reducing sugar (calculated as anhydrous dextrose). Class 2 candies shall

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contain ingredients similar to class 1 except that they shall be colored and flavored according to the requirements of 3.3.1.5.1. In addition, class 2 product centers (plastic filled) candy shall comply with 3.3.1.5.4.1 and the following requirements.

3.3.1.5.1 Colors and flavors. Colors and flavors for all type V candies shall be as shown in table III.

TABLE III. Assortment composition for type V

Class	Color	Flavor	Flavor to be added
1, 2	Orange	Orange	Orange oil (as needed)
1, 2	Yellow	Lemon	Lemon oil (as needed)
1, 2	Light green	Lime	Lime oil (as needed)
2	Medium green	Spearmint	Spearmint oil (as needed)
2	Light red	Clove	Clove oil (as needed)
2	Dark red	Cinnamon	Cassia oil (as needed)
2, 3	White	Peppermint	Peppermint oil (as needed)
2	Light pink	Wintergreen	Wintergreen oil (as needed)
1, 2, 3	Red	Strawberry	Imitation strawberry (as needed)
1, 2, 3	Green	Apple	Imitation apple (as needed)
2	Dark pink	Raspberry	Imitation raspberry (as needed)
2	Medium red	Cherry	Imitation wild cherry (as needed)
1, 2	Yellow	Pineapple	Imitation pineapple (as needed)

3.3.1.5.2 Classes 1 and 3. The pieces shall be of uniform shape and shall be of such size that there shall be not less than 75 nor more than 125 pieces per pound, and the average count shall be not less than 90 pieces per pound. Broken or crushed pieces shall not exceed 2.0 percent by weight.

3.3.1.5.3 Class 1, fruit tablets. Fruit tablets shall comply with 3.3.1.5.1. The assortment before packaging shall consist of equal amounts of five flavors, four of which shall be orange, lemon, lime, and pineapple. There shall be at least four flavors in each bar.

3.3.1.5.4 Class 2, plastic-filled, pulled jackets. Plastic-filled, pulled jacket confections shall comply with the following requirements.

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3.3.1.5.4.1 Centers. The centers of plastic-filled hard candy shall be (a) dark-sweet chocolate, (b) chocolate cream, or (c) confectioners' apple-jam flavored with raspberry, strawberry, orange, lemon, lime, or pineapple. The mix shall consist of (a) or (b), or (a) and (b) combined, and the remainder (c). Any combination of five of the fruit flavor centers under (c) will be acceptable.

3.3.1.5.4.1.1 Moisture. Confectioners' apple-jam centers shall have a maximum moisture content of 24 percent.

3.3.1.5.4.2 Jacket. The jacket shall average not more than 75 percent by weight of the finished piece.

3.3.1.5.4.3 Color and shape. The shapes shall be assorted, standard-filled, hard-candy shapes but shall be not less than six shapes appropriately colored and flavored.

3.3.1.5.4.4 Count. The count per pound for individual shapes shall be not less than 100 nor more than 150, but the mixture shall average not less than 115 per pound.

3.3.1.5.5 Class 3, mint tablets, clear, unsugared. Type V, class 3 shall be clear and unsugared tablets and shall be flavored with not less than 2 1/2 ounces triple-distilled peppermint per 100 pounds.

3.3.1.6 Type VI, sugar, pan-coated confections. Type VI confections shall have a solid, uniform, candy coating applied by the pan-coating method. The coating shall be free from pits (a pit shall be defined as a hole or air bubble that is as great or greater than the head of a common pin), holes, and cracks, and shall be of such hardness that it cannot be easily crushed or cracked. The centers shall be suitably processed before and during the coating process so that at least 98 percent of the product shall withstand all temperature changes encountered below 120°F and 45 percent relative humidity for 24 hours without checking or cracking of the coating. The coated products shall be wax polished in such a manner that minimal seepage of material through the coating occurs and a good glossy sheen will be retained. The pieces of any one class shall be uniform in size and shape. Not more than 2 percent by weight of bulk packaged product shall be broken or crushed.

3.3.1.6.1 Class 1, peanuts. Peanuts conforming to 3.2.27 shall be used. They shall be dry-roasted. They shall then be pan coated by the cold-pan method. The finished product shall be of the "Boston baked bean" type and shall be uniformly medium to dark brown in color.

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3.3.1.6.2 Class 2, jelly beans. Jelly beans shall consist of properly cured starch jelly centers. There shall be a variety of not less than five colors and flavors including the three flavors (orange, lemon, and cherry) specified in 3.3.1.5.1, mixed prior to packaging in equal proportions. The count shall be not less than 160 nor more than 240 per pound. Black jelly beans shall not be included in the assortment.

3.3.1.6.3 Class 3, chocolate disks. Pan-coated chocolate disks shall consist of sweet milk or one of the other standardized chocolates (see 3.2.6). The centers shall be pan coated by the cold-pan method. The pieces shall resemble a double-convex lens in shape. The mixture shall consist of at least five different colors. The count per pound shall not be less than 480 or more than 510. The finished product shall test Salmonella negative in accordance with 4.4.6.

3.3.1.6.4 Class 4, licorice lozenges. The centers of the type VI, class 4 candies shall be licorice candy. The finished product shall be cylindrical in shape. They shall be pan coated by the hot-pan method. There shall be a variety of at least three pastel colors, one of which shall be white, and all shall be anise flavored. The count shall be not less than 300 per pound.

3.3.1.6.5 Class 5, creams. The coating shall be applied by the cold-pan method. There shall be a variety of at least four flavors, one of which shall be a mint. The flavors shall be incorporated into the coatings. Different flavored pieces shall be present in the finished mixture in approximately equal proportions. The pieces shall be suitably colored. The finished pieces shall be disk or cone shaped, and the greatest diameter shall not exceed 5/8 inch.

3.3.1.6.6 Class 6, almonds. The almonds shall be pan coated by the hot-pan method as soon as possible after fumigation and cleaning. There shall be a variety of at least five pastel colors with appropriate flavors. One of the colors shall be white.

3.3.1.6.7 Class 7, bridge mix. Bridge mix shall consist of equal portions by weight of classes 1, 2, 3, 4, and 5. When packaged, a tolerance of not more than 20 percent variation in portions by weight or 10 percent of any flavor or color, of any class within class 7 product, per carton or can will be allowed. Each class shall conform to the respective requirements.

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3.3.1.7 Type VII, enriched sweet chocolate with almonds, buttercrunch, or toffee (bars and disks). The enriched sweet chocolate shall have a texture relatively free from air holes, and the moisture content shall not exceed 1.75 percent by weight. The wrapped bars and disks shall retain their shape and shall not adhere to the wrapper when held for 2 hours at 100°F. The finished product shall test salmonella negative in accordance with 4.4.6. The chocolate portion of the bars and disks shall be produced in accordance with the following formulation:

Ingredients	Percent by weight
Sugar	Not more than 45
Nonfat milk solids	Not less than 17
Added fat (113° ±2°F, Wiley m.p.)	Not less than 21.3
Chocolate liquor (adjusted to 50 percent fat minimum)	Not less than 12
Breakfast cocoa (22-24 percent cocoa butter)	Not less than 3
Sorbitan monostearate	0.5
Polyoxyethylene 20 sorbitan monostearate	0.5
Thiamine mononitrate (vitamin B_1)	1/
Pyridoxine hydrochloride (vitamin B_6)	1/
Ascorbic acid (vitamin C)	1/
Vitamin A	1/
Salt	2 ounces per 100 pounds
Vanillin	1 ounce per 100 pounds

1/ In sufficient quantity to yield the amount specified (see 3.3.1.7.3), in one ounce of the finished product.

3.3.1.7.1 Preparation. The sorbitan monostearate and polyoxyethylene 20 sorbitan monostearate must be melted and mixed with the added fat at a temperature of not less than 150°F, and the chocolate liquor added and mixed at that temperature or higher for a period of at least 30 minutes before the addition of the remaining ingredients. Ninety-nine percent of the finished chocolate product, excluding the almonds, buttercrunch, or toffee, shall pass through a U.S. Standard No. 120 sieve, and 94 percent shall pass through a U.S. Standard No. 325 sieve, when tested as specified in 4.4.7.

3.3.1.7.2 Almonds, buttercrunch, or toffee. The finished bars or disks shall be formulated of 80 percent enriched sweet chocolate and 20 percent roasted diced almonds, buttercrunch, or toffee. The almonds, buttercrunch, or toffee (whichever is used) shall be uniformly distributed throughout the chocolate.

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3.3.1.7.3 Vitamin content. One ounce of the finished product shall contain not less than 0.5 mg per ounce of thiamine mononitrate (vitamin B_1), not less than 0.5 mg per ounce of pyridoxine hydrochloride (vitamin B_6), not less than 20 mg per ounce of ascorbic acid (vitamin C), and not less than 1250 I.U. units of vitamin A.

NOTE: As a suggested guide in manufacturing, losses in processing can be anticipated to be approximately 15 percent for vitamin A, ascorbic acid, pyridoxine hydrochloride, and 30 percent for thiamine mononitrate.

3.3.1.8 Type VIII, fudge bar, chocolate, survival. The following formulation shall be used, except that the individual ingredients may be varied up to 5 percent, more or less, of the specified weight, if necessary, to meet the requirements of the finished bar:

Ingredient	Percent by weight
Sugar, granulated or fondant type	48.0
Corn syrup solids	5.2
Sorbitol solids	10.0
Fat, Domestic (110° ±2°F, Wiley m.p.)	7.3
Chocolate liquor	7.5
Nonfat milk solids	21.3
Salt (see 3.2.29)	0.69
Antioxidant mixture	0.01

3.3.1.8.1 Moisture. Moisture of finished type VIII candy shall be not less than 5 percent nor more than 10.0 percent.

3.3.1.8.2 Finished product. The type VIII product shall be even in texture, not dry or hardened, and shall show no evidence of oil separation. The product shall test salmonella negative in accordance with 4.4.6.

3.3.2 Dimensions (types I through VIII).

3.3.2.1 Bars and pieces. The finished wrapped bars and pieces for operational rations of all classes shall have the dimensions specified in table IV.

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TABLE IV. Dimensions of bars and pieces for operational rations

Type	Bar (wrapped)	Piece
I, II	3 1/8 by 1 1/4 by 3/4 inches (± 3/16 inch)	-
III (flavor 3)	-	3/4 by 3/4 inch square (± 1/32 inch) and 5/16 inch maximum thickness
III (flavor 3, styles 1 and 4 caramels) (see 6.1) 2/	4 1/2 by 3/4 by 1/2 inch 1/ (Three individual pieces laid end to end to form bar)	1 1/2 by 3/4 by 1/2 inches (± 1/32 inch)
IV	-	1 1/2 by 3/4 inches (± 1/32 inch) and 5/16 inch maximum thickness
V	-	3/4 by 3/4 inch square and 3/8 inch maximum thickness
VII	3 1/8 by 1 9/32 by 7/16 inches maximum	-
VIII	Rectangular with rounded corners with a radius of 11/16 inch (± 1/16 inch), having dimensions 3 1/2 by 1 13/16 inches and 5/8 inch maximum thickness	-

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1/ Meal, Ready-to-Eat, Individual. Alternatively, six individual pieces, each measuring 3/4 by 3/4 by 1/2 inch, of type III, flavor 3, style 1 carmels, individually wrapped in accordance with 5.1.1.4.1, laid end to end, and overwrapped to form a bar measuring 4 1/2 inches by 3/4 inch by 1/2 inch may be permitted, when specified for the Meal, Ready-to-Eat Ration provided that the product meets all other applicable portions of this document including the net weight requirement set forth for the type III bar in table V (the net weight requirement per piece in table V, the count per pound requirement (see 3.3.1.3.2) and minor defect 207 in table XII do not apply to this alternative).

2/ Pieces shall be individually wrapped in accordance with 5.1.1.4.1.

3.3.2.2 Disks. All finished wrapped disks, except the finished wrapped disks of type VII, shall not exceed 2 3/4 inches in diameter and 9/16 inch in thickness. When a rectangular wrap is used, the excess material at the corners shall be folded over the edges and against one face of the disk before checking the dimensional limits as specified. The finished wrapped disk of type VII shall not exceed 2 3/4 inches in diameter and 5/16 inch in thickness.

3.3.2.3 Bags. The bags for type VI, class 3 product shall be not more than 6 inches in length and 2 5/8 inches in width.

3.3.2.4 Rolls. There are no dimensional requirements for rolls, except for certain ration items, as specified.

3.3.3 Net weights for operational rations. Net weights of candy for operational rations shall be minimum weight in accordance with table V.

TABLE V. Net weights 1/

Type	Class	Bar	Disk	Roll	Piece	Bag
I	1	1 1/2 oz	1 1/2 oz	--	--	--
	2,3	1 3/4 oz	1 3/4 oz	--	--	--
II	--	2 oz	--	--	--	--

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TABLE V. Net weights 1/ - Continued

Type	Class	Bar	Disk	Roll	Piece	Bag
III	--	--	--	--	--	--
Flavor 1		--	--	1/2 oz	--	--
Flavor 3		1 1/4 oz	--	--	1/5 oz	--
IV	--	--	--	--	1/4 oz 2/ (6.8 grams)	--
V	1,3	1 oz	--	--	2.8 grams	--
VI	3	--	--	--	--	1 1/2 oz
VII	--	1 oz	1 oz	--	--	--
VIII	--	2 oz	--	--	--	--

1/ All net weight requirements shall be applicable to the lot average with a minus 5 percent tolerance per sample unit, except that the net weight requirements shall be applicable to the individual unit for flavor 3 of type III and for types IV, V, VII, and VIII candies.

2/ Packed as 2-piece unit pack

3.4 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110: Current Good Manufacturing Practice in Manufacture, Processing, Packaging or Holding of Human Foods; and the plant sanitation requirements of the appropriate Government inspection agency.

3.5 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

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4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of supervision necessary to assure compliance with the requirements of this document.

4.3 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.3.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.3.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition and other requirements specified in 3.2 (except for component examination and ingredients testing as specified in 4.3.1.1.1 and 4.3.1.1.2) shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be evident by examination of pertinient labels, markings, US Grade Certificates, certificates of analyses or other such valid documents acceptable to the inspection agency. In addition, prior to use, each ingredient shall be examined organoleptically, as necessary, to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.3.1.1.1 Examination of almonds for count per ounce. Almonds to be used for type VI, class 6 confection shall be examined for count to determine conformance to 3.2.2. One ounce of almonds shall be extracted from each of five containers and counted. If the average of the total determinations is more than 26, the almonds shall be in nonconformance with the requirement (see 3.2.2).

4.3.1.1.2 Testing of ingredients. Criteria pertinent to testing of ingredients shall be as shown in table VI. Requirements shall be applicable to the lot average. The sample of ingredients for testing shall be a composite derived from five primary containers, or from all primary containers in a lot in the event a lot consists of less than five primary containers. Test procedures shall be as specified in 4.4 and, in addition for certain characteristics, the pertinent requirement paragraph. Test results shall be reported to the same unit or decimal place specified for the specific requirement, except that test results for flavor, odor, and color of vegetable fat shall be reported as pass or fail. An ingredient not conforming to one or more test requirements shall be cause for rejection of the ingredient, lot or any finished product made therefrom.

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TABLE VI. Ingredient tests

Ingredient	Sample quantity	Characteristics	Requirement paragraph
Almond pieces (unroasted)	1 pound	Sieve size, free fatty acids	3.2.2
Cocoa (medium fat)	4 ounces	Fat	3.2.8
Coconut	8 ounces	Moisture, free fatty acids, pH	3.2.9
Corn syrup	8 fluid ounces	Reducing sugars	3.2.13
Vegetable fat w/o antioxidants	1 pound	Free fatty acids	3.2.19
Vegetable fat w/antioxidants	1 pound	Free fatty acids, stability, odor, color, flavor, moisture and volatile matter, melting point	3.2.19
Salt	8 ounces	Copper, iron	3.2.29
Invert sugar, liquid sugar, granulated sugar, fondant sugar	8 avoirdupois or fluid ounces, as applicable	Ash	3.2.35, 3.2.36, 3.2.41
Buttercrunch	1 pound	Sieve size	3.2.40
Toffee	1 pound	Sieve size	3.2.44

4.3.1.1.3 Material certification. Material listed below shall be accepted on the basis of contractor's certificate of conformance to the indicated requirements:

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Material	Requirement paragraph
Aluminum	5.1.1.1.2a and 5.1.1.2.2
Bags	5.1.1.5
Cans	5.1.1.6
Disk wrapper	5.1.1.2.1 and 5.1.1.2.2
Envelope material	5.1.1.1.3
Laminated foil	5.1.1.1.2b
Paper, glassine, grease proof	5.1.1.1.1b, 5.1.1.1.2, and 5.1.1.2.1
Paper, glassine or sulfite waxed	5.1.1.3
Paper, kraft or sulfite waxed	5.1.1.6
Paperboard	5.1.1.1.1
Polypropylene bags	5.1.1.2.2c

4.3.2 In-process examinations.

4.3.2.1 Formulation, procedures, and controls. Examination shall be performed to determine conformance to formulation requirements of types I, II, III, IV, V, and VII products and procedures and control requirements applicable to type I product. Records of formulations and records reflecting time and temperature controls associated with the required procedure shall be maintained. When specified (see 6.1), the Government inspector shall perform continuous examination of actual formulations or procedures and controls or both. Nonconformance shall be cause for rejection of the involved product.

4.3.2.2 Weight of coatings. An examination shall be performed during the manufacture of type I product to determine conformance to requirements for weight of coating. Criteria pertinent to the examination shall be as follows:

a. Prior to coating of type I product, randomly select 50 units.
b. Weigh the units and record total weight.
c. After the product has been coated, randomly select 50 units and record total weight. Calculations shall be as follows:

Percent by weight of coating

$$= \frac{\text{Weight of product after coating minus weight of product before coating}}{\text{Weight of product after coating}} \times 100$$

One sample determination shall be made from each lot and shall be applicable to each class of product. A percentage less than that required (see 3.3.1.1) shall be cause for rejection of the lot.

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4.3.3 Examination of end item.

4.3.3.1 Examination of paperboard carton. Cartons shall be examined for the defects in table VII. The lot size shall be expressed in cartons. The sample unit shall be one filled and closed carton, either wrapped or unwrapped, as applicable. The inspection level shall be S-3, and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 10.0 for total defects.

TABLE VII. Paperboard carton defects

Category		Defects
Major	Minor	
101		Overwrap missing when required
102		Overwrap not one of materials specified
	201	Overwrap not completely sealed or has tear
	202	Level A carton not closed as required
103		Level A carton has tear or hole
104		Not clean
105		Prominent odor not associated with product
106		Nomenclature missing, incorrect or illegible
	203	Nomenclature on nonration items not at least 1 inch high
	204	Other required label information missing, incorrect or illegible

4.3.3.2 Examination of intermediate packaging. The intermediate packaging shall be examined for the defects in table VIII. The lot size shall be expressed in cartons or cans, as applicable. The sample unit shall be one filled paperboard carton or one can, as applicable. The inspection level shall be S-3, and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 10.0 for total defects.

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TABLE VIII. Intermediate packaging defects

Category		Defects
Major	Minor	
101		Liners (when required) missing or not completely protecting product from container
102		Liner not type of material required
103		Liners have prominent odor not associated with product
104		Liners not clean
105		Number of bars, disks, rolls or bags not as specified or marked, as applicable
106		Number of units per bar for type II, III, or V bars not as required 1/
	201	Flavor variety of type II or V bar not as required 1/
	202	More than 10 percent variation from required weight for any flavor or color, as applicable, for bulk packaged pieces (type II, III, V, and VI) 2/
	203	Not six different shapes in type V, class 2 product
	204	Colors of type VI, class 3, product in a bag not as required 1/
	205	More than 10 percent variation from required weight of any flavor or color, as applicable for any class within class 7 of type VI product 2/
107		More than 20 percent variation from required weight for any class within class 7 of type VI product

1/ One bar or bag, as applicable shall be examined from a sample unit.

2/ Defect is applicable when requirement is for equal or approximate equal weights of color or flavors, as applicable.

4.3.3.3 Examination of primary packaging. The primary packaging shall be examined for the defects in table IX. The lot size shall be expressed in terms of the sample unit except for bulk packaged pieces, which shall be expressed in terms of pounds. The sample unit shall be one bar, roll, disk, bag, or wrapped piece, as applicable. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total defects. For meal-ready-to-eat procurements, the AQL shall be 0.65 for major defects and 4.0 for total defects.

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TABLE IX. Primary packaging defects

Category		Defects	Bar, roll, disk	Bag	Wrapped piece 1/
Major	Minor				
101		Packaging material(s) not as specified	X	X	X
102		Package not fabricated, sealed or closed in the manner required, as applicable	X	X	X
103		Tear, hole, or open seal 3/16 inch or larger	X	X	X
	201	Tear, hole, or open seal less than 3/16 inch	X	X	X
104		Delamination in laminated envelope (see 5.1.1.1.3)	X		
105		Size of package not as specified (see 3.3.2) 2/	X	X	
	202	Not a conforming wrap	X		
	203	Overlap of wrap less than 1/8 inch (see 5.1.1.1.1) 2/	X		
106		Prominent odor not associated with product	X	X	X
107		Not clean	X	X	X
108		Nomenclature missing, incorrect or illegible	X	X	
	204	Other required label information missing, incorrect or illegible	X	X	
109		Crushed or misshapen package	X		

1/ For type V bars, examine one wrapped piece from a bar.

2/ Measuring device shall be calibrated in 1/32-inch increments.

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4.3.3.4 Examination of net weights. Examination shall be made of the net weights as specified in table X. The lot size shall be expressed in terms of the sample unit. The sample unit shall be one bar, disk, roll, or bag or bulk packaged can or carton, as applicable. The inspection level shall be S-3, and the AQL shall be 1.0 for major defects and 2.5 for minor defects, expressed in terms of defects per hundred units.

TABLE X. Net weights defects 1/ 2/

Category		Defects
Major	Minor	
	201	Type I, III, or VI (bag) more than 5 percent under applicable net weight specified in tabulated data of table V
101		Type II (bar), IV (2-piece unit pack), V (tablet from a bar), VII (bar or disk), or VIII less than applicable net weight specified in tabulated data of table V
	202	Net weight of bulk packaged pieces more than 2 percent under labeled net weight

1/ Report results to nearest 0.1 gram for each unit examined under defect 201 and nearest 0.25 gram for each unit examined under defect 202. The weighing instrument shall be calibrated in 0.1 gram for weighing type II and VIII units and 0.1 gram for weighing type IV and V units.

2/ Lot shall be rejected if sample data indicate a lot average net weight less than required net weight for defects 201 and 202.

4.3.3.5 Examination for leakage and delamination. Type VIII bars shall be examined for the defects in table XI (see 4.4.10 for procedure). The lot size shall be expressed in terms of bars. The sample unit shall be one type VIII bar. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.0.

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TABLE XI. Type VIII leakage and delamination defects

Category	Defects
Major	
101	Leakage
102	Delamination 1/

1/ Examine after removal from water

4.3.3.6 Examination for product characteristics. The finished product shall be examined for the defects in table XII. The lot size shall be expressed in terms of the sample unit. The sample unit shall be one bar, disk, roll, bag, or bulk packaged can or carton. The inspection level shall be S-2, and the AQL, expressed in terms of defects per hundred units, shall be 1.0 for major defects and 10.0 for total defects.

TABLE XII. Product characteristics defects 1/

Category		Quantity to examine per sample unit 2/	Defects
Major	Minor		
			Appearance and condition all types
101		1 bar, disk, roll, or piece	Wrapper on product does not come off intact (wrapper or portion thereof strongly adheres to product)

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TABLE XII. Product characteristics defects 1/ - Continued

Category		Quantity to examine per sample unit 2/	Defects
Major	Minor		
102		1 bar, disk, roll, or piece	Appearance of product indicates off condition (moldy, dried out, etc.)
103		1 bar, roll, or disk	Broken or crushed bar, roll, or disk
104		Contents of 1 carton or can	More than 2 percent by weight of bulk packaged pieces (except type III) broken or crushed
			Type I
	201	1 bar or disk	Coating has abnormal irregularities, such as but not limited to cracks, chips, rough spots
105		1 bar or disk	Portion of center protrudes through coating
	202	1 bar or disk	Coating not smooth
	203	1 bar or disk	Coating gritty
			Type II
	204	1 bar or 4 pieces from a bulk package	Not uniform in shape or size
106		1 piece of each flavor	Color does not represent applicable flavors
	205	1 pound of pieces	Number of pieces (bulk packaged) in a pound less than 45 or more than 65
			Type III
			Appearance and condition of all type III flavors
	206	1 roll	Roll not lightly scored to form 5 distinct sections
107		1 roll	Outer surface of roll grained
108		1 roll	Inner portions of roll not grainy
109		1 roll or 1 piece	Not chewy in character

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TABLE XII. Product characteristics defects 1/ - Continued

Category		Quantity to examine per sample unit 2/	Defects
Major	Minor		
	207	1 pound of bulk packaged pieces from a carton	Less than 55 or more than 90 pieces
			Type IV
110		2 pieces	Not white
			Type V
	208	1 piece	Class 2 product not uniform in shape for any flavor
111		1 bar or 1 piece of each color	Colors do not correspond to flavors
	209	1 of each jelly flavor	Class 2 jackets do not match flavor of jelly filled centers
	210	1 pound	Class 2 product has less than 100 or more than 150 pieces in 1 pound 3/
	211	1 pound	Class 1 and class 3 product has less than 75 or more than 125 pieces in 1 pound 4/
			Type VI 5/
	212	1 ounce	Count per pound (calculated from 1 oz) not as specified for classes 2, 3, and 4
	213	1 1/2 ounces	Count per pound (calculated from 1 1/2 oz) not as specified for class 3
			Appearance and condition of all type VI classes
112		1 piece of each class	Surface not free of pits, holes or cracks, as qualified in 3.3.1.6
113		1 piece of each class	Piece can be crushed or cracked by light finger pressure
114		1 piece of each class except classes 1 and 6	More than minimal seepage of interior material through coating

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TABLE XII. Product characteristics defects 1/ - Continued

Category		Quantity to examine per sample unit 2/	Defects
Major	Minor		
	214	Measure 1 piece of class 5 and examine piece of each class for shape	Pieces of any one class not shape or size specified, or not uniform in size or shape 6/
	215	1 piece	Class 1 product not medium to dark brown
115		1 piece	Class 1 product not "Boston baked bean" type
116			Class 2 colors do not represent flavors
117		Content of 1 bag	More than 1 broken or crushed piece in a bag
			Type VII
	216	1 bar or disk	Texture not relatively free of air holes
	217	1 bar or disk	Almond, buttercrunch, or toffee pieces not uniformly distributed through bar or disk
			Type VIII
118		1 bar or disk	Not an even texture
119		1 bar or disk	Texture hard or dry
120		1 bar or disk	Evidence of oil separation
			Odor and palatability of all type VIII styles
121		1 unit of each class and flavor, as applicable	Objectionable odor
122		1 unit of each class and flavor, as applicable	Not equal in palatability to bid sample 7/

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TABLE XII. Product characteristics defects 1/ - Continued

Category		Quantity to examine per sample unit 2/	Defects
Major	Minor		
123		1 unit of each class and flavor, as applicable	Flavor objectionable or not typical of product 8/

1/ Presence of foreign material (e.g. dirt, glass, metal, filth, insects, paint) shall be cause for rejection of the lot.

2/ Where feasible, the same unit of examination may be utilized for more than one defect characteristic.

3/ The lot shall be rejected if average count is less than 115 pieces per pound.

4/ The lot shall be rejected if average count is less than 90 pieces per pound.

5/ When examining class 7, defect characteristics for classes 1 through 5 shall apply.

6/ Measuring device shall be calibrated in 1/16-inch increments.

7/ Applicable only when bid sample is required.

8/ Not applicable when bid sample is required.

4.3.3.7 Examination of cans. Examination of cans for condition and seam leakage shall be in accordance with the applicable quality assurance provisions of PPP-C-29.

4.3.3.8 Examination of can coating. When cans are required to be coated in accordance with TT-C-495, examination of coating shall be performed in accordance with that document.

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4.3.3.9 Examination of can labeling. Examination of can labeling shall be in accordance with the quality assurance provisions of MIL-L-1497.

4.3.4 Testing of finished product.

4.3.4.1 Analytical. Applicability of analytical tests to types and classes of product shall be in accordance with the checked characteristics in table XIII. Test procedures shall be in accordance with 4.4. The sample for testing shall be approximately a 1-pound composite of each type and class of product derived from eight containers. Lot size shall be expressed in terms of the package for bars, rolls, disks or bags and in terms of weight for bulk packaged pieces. Results shall be reported to the same unit or decimal place as specified for the applicable requirement except the test for salmonella shall be reported as pass or fail. A test result not in conformance with the pertinent requirement shall be cause for rejection of the lot.

4.3.4.2 Physical. Applicability of physical tests to types of product shall be in accordance with the checked characteristics in table XIII. Test procedures for wrapper adherence, temperature resistance, and "sweat" shall be in accordance with the pertinent requirement paragraphs. Test procedure for particle size shall be in accordance with 4.4. The sample unit for the wrapper adherence and "sweat" test shall be one wrapped bar or disk, as applicable. The sample unit for temperature resistance shall be 50 pieces from a container for each class of type VI product, including each class within class 7. The sample size shall be eight. The sample for particle size test shall be a representative eight bars or disks, as applicable. The chocolate shall be composited after grating. Lot size shall be expressed in terms of disks, bars, or bags; or, in the case of bulk packaged or packed pieces, in terms of pounds. A test result not in conformance with the pertinent requirements shall be cause for rejection of the lot.

4.3.5 Examination of waterproof case liners. The filled and closed waterproof case liner shall be examined in accordance with the quality assurance provisions in the appendix of MIL-L-10547.

4.3.6 Examination of fiberboard containers. Examination of filled, closed, and reinforced fiberboard containers shall be in accordance with the quality assurance provisions in the appendix of PPP-B-636. In addition, the following defects shall be included in the examination:

a. Major - Required national stock number, item description, contract number, or date of pack markings missing, incomplete, or illegible.

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b. Major - Protective pad and tape, when required, missing, not as specified, or not completely covering stitches.

c. Minor - Number of cans or paperboard cartons not as marked.

d. Minor - Marked net weight exceeds that specified.

e. Minor - Other required markings missing, incorrect or illegible.

f. For commercial packing, only the defects pertinent to marking and number or weight of contents shall apply.

4.3.7 Examination of unit loads. Unit loads shall be examined in accordance with the quality assurance provisions of MIL-L-35078 and the applicable specification sheet for the type and class of load specified.

4.4 Methods of inspection.

4.4.1 Chemical analyses and physical tests. Chemical analyses and physical tests shall be made in accordance with the methods in table XIV from Official Methods of Analysis of the Association of Official Analytical Chemists (AOAC), the Official and Tentative Methods of the American Oil Chemists Society (AOCS) or other methods contained herein, whichever is applicable.

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TABLE XIII. Analytical and physical tests

Type	Moisture	Reducing sugars	Almond pieces	pH	Vitamins	Salmonella	Wrapper adherence and shape	"Sweat"	Particle size	Cracking and checking
I	X (centers only)					X	X (wrapper adherence only)			
II	X			X				X		
III	X						X (wrapper adherence only)			
IV	X									
V	X (class 1, 2 apple-jam centers and 3)	X (class 1 and 3)								
VI						X (class 3)				X
VII	X		X		X	X	X		X (chocolate only)	
VIII	X					X				

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TABLE XIV. Test methods

Test and ingredients/product	Source	Method
Ingredients:		
Almonds, diced roasted		
Particle size	MIL-C-10928	3.2.2 & 4.4.8
Free fatty acids (oil extraction - clean fraction of cold press)	AOCS	C_a 5_a -40
Cocoa powder - medium fat cocoa		
Fat	AOAC Cocoa Bean and Its Products	Fat - Method - I
Coconut, tenderized, moistened		
Moisture	AOAC Vegetable Products, Processed	Karl Fischer
Free fatty acids	AOCS	
	Oil Extraction	A_a 6-38
	Free Fatty Acids	C_a 5_a -40
pH	AOAC Cereal Foods; Wheat Flour	Hydrogen ion concentration - electrometric
Fat, vegetable, hydrogenated		
Stability	AOCS	Cd 12-57
Flavor, odor, & color	MIL-C-10928	4.3.1.1.2
Free fatty acids	AOCS	C_a 5_a -40
Moisture (volatile matter)	AOCS	C_a 2d-25 Vacuum oven
Wiley melting point	AOCS	C_c 2-38

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TABLE XIV. Test methods - Continued

Test and ingredients/product	Source	Method
Ingredients:		
Salt Copper & iron	AOAC Animal Feed; Minerals in Feeds by Atomic Absorption Spectrophotometry	AAS
Salt Copper & iron Continued	Absorption Spectro-photometry	
Sugar, granulated, powdered, liquid or invert and fondant sugar Ash	AOAC Sugars & Sugar Products; Sugars & Sirups	Ash - Method I
Finished Product or Components:		
Type I coconut candy center Moisture	AOAC Sugars & Sugar Products; Sugars & Sirups	Drying on quartz sand
Type I chocolate or vanilla fudge centers Moisture	Same as above	Same as above
Types I, VI (class 3 only) VII, VIII Salmonella	MIL-C-10928	4.4.6

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TABLE XIV. Test methods - Continued

Test and ingredients/product	Source	Method
Finished Product or Components:		
Type I		
Wrapper adherence	MIL-C-10928	3.3.1.1
Type II, style I		
Sweat - wrapper	MIL-C-10928	3.3.1.2
Type II, style I		
Moisture	MIL-C-10928	4.4.2
pH	AOAC Cereal Foods; Wheat Flour	Hydrogen ion concentration
Type III		
Moisture	AOAC Sugars & Sugar Products; Sugars and Sirups	Vacuum drying on quartz sand
Wrapper adherence	MIL-C-10928	3.3.1.3.2
Type IV		
Moisture	AOAC Sugars & Sugar Products; Sugars & Sirups	Vacuum drying
Type V, classes 1 & 3		
Moisture	MIL-C-10928	4.4.3
Reducing sugar (dextrose)	AOAC Sugars & Sugar Products; Sugars & Sirups	Dextrose

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TABLE XIV. Test methods - Continued

Test and ingredients/product	Source	Method
Finished Product or Components:		
Type VI		
Checking and cracking	MIL-C-10928	3.3.1.6
Type VII		
Moisture	AOAC Cacao Bean and Its Products	Moisture
Wrapper adherence	MIL-C-10928	3.3.1.7
Particle size	MIL-C-10928	4.4.7
Vitamins	MIL-C-10928	4.4.4
Almonds	MIL-C-10928	4.4.5
Type VIII		
Moisture	AOAC Sugars & Sugar Products; Sugars & Sirups	Drying on quartz sand

4.4.2 Moisture in type II starch jelly candies. Moisture in type II starch jelly candies shall be determined as follows:

a. Scoop small portion of soft material from inside of starch jelly candy.
b. Place immediately on prism of refractometer.
c. Read refractive index.
d. Make necessary temperature correction to 68°F. (See table 43,009, Methods of Analysis of the AOAC)
e. Reading is percent solids as percent sucrose.

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f. Calculation:
Percentage of solids by corrected refractive index minus 0.5 empirical factor = the percentage of solids in candy.
100 percent minus the percentage of solids in candy = the percent moisture.

g. Clean prism thoroughly with distilled water and dry.

4.4.3 Moisture in type V, hard candy. Moisture in type V, hard candy, shall be determined as follows: Pulverize a representative sample with a large mortar and pestle, working rapidly to avoid absorption of atmospheric water vapor. Pass through a U.S. Standard No. 100 sieve, and transfer to a jar, closing tightly. Quickly weigh 3 to 5 grams of the prepared sample into a conditioned, weighed moisture dish with cover (about 55 millimeters (mm) diameter, 20 mm high). Dry at 158°F under pressure not exceeding 50 mm mercury for 16 hours. Cool in a desiccator and reweigh. Loss of weight represents weight of water in sample.

4.4.4 Vitamin determination. Sample of type VII chocolate shall be taken from the same lot before and after fortifying with vitamins. The amount of ascorbic acid, thiamine mononitrate, pyridoxine, and vitamin A in the fortified chocolate shall be calculated by difference in the assay values of the unfortified and fortified samples. The vitamins shall be determined by the following methods described in Methods of Vitamin Assay of the Association of Vitamin Chemists:

Ascorbic acid
Thiamine mononitrate
Vitamin A
Pyridoxine

4.4.5 Almond determination. Five ounces of sample product shall be broken up and placed in a 1-liter beaker containing 700 mL of water. The beaker and contents shall be heated to 203°F and stirred for 15 minutes using a heated magnetic stirrer or its equivalent. The almond pieces shall be allowed to settle and shall be poured onto a tared boat-shaped screen. The screen, measuring approximately 3 1/2 by 2 1/2 inches with sides approximately 3/4-inch high, may be constructed from a U.S. 32 mesh stainless steel with 0.010-inch wire diameter, 0.54 millimeter width of opening (46.5 percent open area). The collected almond pieces shall be washed with hot water to remove any trace of chocolate. The almonds contained in the tared screen boat shall be transferred onto a tared aluminum dish and dried to a constant weight in a vacuum oven for at least 16 hours at 158°F. The weight of almond pieces shall be determined by weighing and subtracting the tare weights of both the boat and aluminum dish. The percent almond pieces shall be calculated by dividing

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the weight of product assayed initially (circa 142 grams) into the weight of dried almond pieces recovered after drying. This value multiplied by 100 equals percent by weight almonds recovered.

4.4.6 Microbiological-Salmonella. Salmonella test shall be the FDA Qualitative Procedure Assay listed in the FDA Bacteriological Analytical Manual or latest approved FDA test.

4.4.7 Particle size determination for type VII products. Particle size determination shall be made as follows:

a. Freeze hard, sufficient sweet chocolate and separate from almonds, buttercrunch, or toffee (type VII) to meet requirement in (d).
b. Grate the material.
c. Screen particles through a U.S. Standard No. 20 sieve.
d. Weigh not less than 50 grams into tared laboratory bottles.
e. Add petroleum ether to cover chocolate.
f. Hold in stoppered bottles for 1 hour, shaking well occasionally.
g. Wash dissolved chocolate through clean, tared, U.S. Standard No. 120 and No. 325 sieves until the ether passing through sieves is clear.

4.4.8 Particle size of almond, buttercrunch, or toffee pieces. Shake almond, buttercrunch, or toffee pieces on sieves specified in 3.2.2, 3.2.40, or 3.2.44, respectively, until no more pieces pass through. Weigh amounts retained on and passing through specified sieves.

4.4.9 Paper tests. Paper shall be tested in accordance with method 5022 for basis weight and method 3017 for turpentine test of FED-STD-101, and in accordance with ASTM D 87 for the melting point of paraffin wax.

4.4.10 Envelope seal examination procedure. Envelope seals shall be examined as follows: Submerge the envelope in water in a vacuum desiccator, Mead tester, or equivalent device. Draw 5 inches of vacuum and hold for at least 10 seconds. Observe envelope (a) for air leakage in large quantities from any point, (b) for a fine stream of bubbles anywhere in the seal area, and (c) for failure to inflate. Envelopes that do not inflate under vacuum shall be opened and examined for water penetration. Leakage of air under the first two conditions or wetting of the product under the last condition indicates an unacceptable envelope.

4.4.11 Tin coating weights. Tin coating weights shall be determined by any method specified in PPP-C-96.

4.4.12 Polyethylene and polypropylene thickness test. Thickness shall be tested in accordance with L-P-378.

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4.4.13 Aluminum foil. Thickness of aluminum foil shall be determined in accordance with QQ-A-1876.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Bars.

5.1.1.1.1 Type I and type III. Each type I and type III bar shall be completely wrapped in one of the following materials:

a. Cellophane complying with type IV, class 2 of L-C-110.
b. Glassine or greaseproof paper having a minimum turpentine test of 600 seconds when tested in accordance with 4.4.9.
c. Bi-axially oriented polypropylene minimum 150 gage with patterned cold seal cohesive. The cohesive shall be suitably formulated for contact with food and shall not impart any odor or flavor to the product being packed.

The wrap shall overlap itself a minimum of 1/8 inch and shall be held in place with a body seal or an end seal effected by gluing or heat sealing or by crimping the end. A tuck-end wrap may be used provided the tucked ends are held securely in place by the body seal. Prior to being wrapped, the bar may be placed in a tray, card, or boat made of paperboard consisting of not more than 50 percent ground wood pulp and the remainder consisting of sulfate or sulfite pulp or a combination of the two. The paperboard may be colored. The size of the packaged bar shall be as specified in 3.3.2.1.

5.1.1.1.2 Type VII. The type VII candy bar shall be unit packed in accordance with 5.1.1.1.1. Alternatively, the following materials may be used for wrappers:

a. A sheet of dry finish aluminum not less than 0.00065-inch thick (see 4.4.13).

b. A sheet of 0.00035-inch thick aluminum foil laminated with 5 pounds of wax per ream (24 by 36 - 500) to glassine paper having a minimum basis weight of 24 pounds per ream (24 by 36 - 500). The size of the unit packed bar shall be as specified in 3.3.2.1.

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5.1.1.1.3 Type VIII. The type VIII candy bar shall be unit packed in a snug-fitting envelope formed from one sheet of material folded and heat sealed to form a tube, followed by heat sealing of the top and bottom edges. The envelope material shall be made by hot-extruding or laminating not less than 0.001-inch thick, nor more then 0.002-inch thick polyethylene on 0.00035-inch thick aluminum foil ± 10 percent, and laminating the plain side of the foil to 0.0005-inch thick polyester ± 20 percent. The laminating agent shall be an odorless, permanently plastic, water-resistant material other than wax. There shall be no evidence of delamination or leakage when examined visually after fabrication of envelopes and heat sealing. The dimensions of the packaged bar shall be as specified in 3.3.2.1.

5.1.1.2 Disks.

5.1.1.2.1 Type I. Type I disks shall be unit packed in a snug-fitting rectangular wrap made of plasticized, grease-proof glassine having a minimum basis weight of 25 pounds per ream (24 by 36 - 500) coated with a minimum 10 pounds per ream (24 by 36 - 500) of low density polyethylene. The filled unit pack shall be closed by heat sealing. The dimensions of the packaged disk shall be as specified in 3.3.2.2.

5.1.1.2.2 Type VII. The type VII disks shall be wrapped in a material as specified below. The dimensions of the wrapped disks shall be as specified in 3.3.2.2.

a. A sheet of 0.00065-inch thick dry finish aluminum foil. The wrapper shall be of the conforming type and shall be closed by pressing the ends securely against the disk.

b. A wrap of plasticized, grease-proof glassine having a minimum basis weight of 25 pounds per ream (24 by 36 - 500) coated with a minimum 10 pounds per ream (24 by 36 - 500) of low density polyethylene. The wrap shall be heat sealed.

c. A wrap of 90 gage, polymer-coated, bi-axially oriented polypropylene. The wrap shall be heat sealed.

5.1.1.3 Rolls, type III. One roll of type III shall be wrapped in a sheet of waxed glassine or waxed sulfite paper having a minimum total basis weight of 27 pounds per ream (24 by 36 - 500). The wax content of the finished sheet shall be not less than 5 pounds per ream (24 by 36 - 500). Prior to being wrapped, the roll may be placed on a paper board tray, card, or boat, as specified in 5.1.1.1.1.

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5.1.1.4 Pieces.

5.1.1.4.1 Type III, type IV, and type V, classes 1 and 3. Types III and V (classes 1 and 3) candies shall be individually wrapped in a close conforming wrap made of cellophane complying with type IV, class 2 of L-C-110. Two pieces of type IV candy, placed edge to edge, shall be wrapped in the same material as specified for types III and V candies. The size of wrapped candies shall be as specified in 3.3.2.1.

5.1.1.4.2 Bar arrangement.

5.1.1.4.2.1 Type II. Four pieces of type II candy shall be unit packed in accordance with 5.1.1.1.1, except that only the specified cellophane shall be used. The size of the unit packed bar shall be as specified in 3.3.2.1.

5.1.1.4.2.2 Type V. Ten to 14 pieces of the individually wrapped type V (class 1 or 3) candy, arranged on edge, face to face, shall be tightly wrapped and sealed in accordance with 5.1.2. For use in the Food Packet Survival Abandon Ship (MIL-F-16895): Ten pieces of individually wrapped type V, class 1 candy arranged on edge, face to face, shall be tightly wrapped and sealed.

5.1.1.4.2.3 Type III. When type III is specified for Meal, Ready-to-Eat (see 6.1), three or six individually wrapped pieces shall be laid end-to-end and overwrapped in accordance with 5.1.1.1.1.

5.1.1.5 Bags. A minimum of 1 1/2 ounces of type VI, class 3 product shall be filled into a bag having dimensions as specified in 3.3.2.3. The bag shall be fabricated with nitrocellulose coated pouch paper having a minimum basis weight of 28 pounds per ream. Alternatively, high density polyethylene coated paper may be used. The polyethylene laminate shall have a thickness of not less than 0.002 inch. The filled paper/polyethylene bag shall be closed by cold seal. The filled paper/nitrocellulose bag shall be closed by heat seal.

5.1.1.6 Cans. A lot average of 1 to 5 pounds of candy pieces, as applicable, of the type and class specified with not more than 2 percent tolerance per individual sample unit, shall be unit packed in an open-top style, round, metal can with soldered or welded side seam and compound lined, double-seamed ends. Type V, class 2 candy pieces shall be individually wrapped as specified in 5.1.1.4.1. The can shall be made throughout from commercial 0.25-pound electrolytic tin plate per base box (see 4.4.11), and shall be coated overall on the outside with a coating conforming to type I of TT-C-495. For other than type V, class 2 candy pieces, the can shall be furnished with a full height liner and top and bottom disks of bleached kraft or sulfite paper having a minimum basis weight of 20 pounds per ream (24 by 36 - 500), and wet waxed overall with not less than 9 pounds per ream of a fully refined paraffin wax having a melting point of not less than 125°F. The

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filled and sealed can shall not leak when examined in accordance with 4.3.3.7. When specified (see 6.1), 3 pounds of individually wrapped pieces consisting of 25 percent of type III, style 4, flavor 2; 50 percent of type V, class 2, flavor 2; and 25 percent of type V, class 1, flavor 1 shall be unit packed in a size 603 by 700 metal can as specified above.

5.1.1.7 Paperboard carton (pieces). A lot average of not less than 5 pounds, with not more than -2 percent tolerance per sample unit of candy pieces, shall be unit packed in a full telescope style, setup box complying with type I of PPP-B-676 or a folding carton complying with style II of PPP-B-566. The full telescope box shall be secured with a strip of pressure-sensitive tape applied over each side and the bottom of the box. Boxes shall be constructed of white patented coated newsboard, solid sulfate board, or solid sulfite board with a minimum thickness of 0.032 inch. The newsboard shall have the coating on the inside.

5.1.2 Commercial.

5.1.2.1 Unit packs. The product shall be unit packed to afford adequate protection against deterioration, physical damage, and contamination during shipment from the supply source to the first receiving activity. The contractor may use his standard practice when it meets these requirements.

5.1.2.2 Paperboard cartons and wrappers (types I, II, III, V, VII, and VIII, bars or rolls, as applicable, and types I, VII, and VIII). Forty-eight wrapped bars or rolls or 48 disks of one type only shall be placed in a paperboard carton normally used for the product. Alternatively, 50 wrapped bars of type III, flavor 3 shall be placed in a tray and two trays in each shipping carton. Cartons of type V candy shall be enclosed in a completely sealed wrapper of cellophane complying with type IV, class 2 or type VII, class 2 of L-C-110. The other types of candy shall be wrapped with cellophane specified above or with a sheet of bleached kraft or sulfite paper having a minimum basis weight of 35 pounds per ream (24 by 36 - 500), wet waxed overall with not less than 14 pounds per ream of fully refined paraffin wax having a melting point of not less than 125°F.

5.2. Packing. Packing shall be level A, B, or Commercial, as specified (see 6.1).

5.2.1 Level A packing.

5.2.1.1 Cans. Not more than 45 pounds of candy in cans shall be packed in a fiberboard box, constructed, closed, and reinforced in accordance with style RSC, V2s of PPP-B-636.

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5.2.1.2 Unit packs other than cans. Not more than 45 pounds of candy in unit packs other than cans shall be packed in a fiberboard box complying with style RSC, V2s of PPP-B-636. Prior to packing, the candy shall be placed in a nontest corrugated fiberboard box that shall be placed in the shipping container. The shipping container shall be provided with a waterproof case liner, fabricated and closed in accordance with MIL-L-10547 for subsistence items. If flaps of the shipping container are closed by stitching, the case liner shall be protected from the stitches by a paperboard pad placed between the liner and the stitches. The case liner shall be protected from stitches in the manufacturer's joint by a strip of pressure-sensitive adhesive tape placed over the stitches.

5.2.2 Level B packing. The product shall be packed in accordance with 5.2.1.1 or 5.2.1.2, except that the shipping container shall be constructed of fiberboard complying with V3c, V3s, or V4s of PPP-B-636. The shipping container shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive, filament reinforced tape in accordance with the appendix of PPP-B-636, except that two reinforcing bands may be used, one lengthwise and one girthwise.

5.2.3 Commercial packing. Not more than 45 pounds of the product shall be packed in accordance with method II as specified in the appendix of PPP-B-636. The product shall be packed in a manner to ensure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Each container that is packed with paperboard cartons that are not overwrapped with cellophane or waxed paper shall be provided with a polyethylene bag liner having a minimum thickness of 0.002 inch. The bag dimension shall be of sufficient length to permit a slack fill and twist tie closure. The closure shall be secured with a commercial plastic or paper-covered wire tie. All excess air shall be removed from the bag prior to closure. (See 6.2.)

5.3 Unit loads. When specified (see 6.1), the product, packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified. When unit loads are strapped, the strapping shall be limited to nonmetallic strapping.

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5.4 Labeling and marking.

5.4.1 Unit packs.

5.4.1.1 Bars, rolls, disks, and bags. Except for type VIII bar, commercial labeling will be acceptable for domestic and overseas shipments, provided that the following information is included:

Name of product (in letters larger than any others used for the label).
Ingredient statement.
Net weight.
Name and address of manufacturer.
Any other information required by the Federal Food, Drug, and Cosmetic Act, and regulations promulgated thereunder.

5.4.1.1.1 Chocolate fudge surival bar, type VIII. The envelopes shall have the following information clearly and legibly printed in black:

CHOCOLATE-FUDGE BAR
(In letters larger than others used for the label)

For variety, the bar may be crumbled in 8 ounces of boiling water to make a cocoa beverage.

Any other information required by the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

5.4.1.2 Cans. Information specified in 5.4.1.1 shall be labeled on one end or on the body of the can in accordance with MIL-L-1497.

5.4.1.3 Paperboard cartons and wrappers for bars, disks, pieces or bags. The carton wrapper or the carton (where no wrapper is required) for these items shall be labeled with the information required by the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder. The name of product shall be labeled in letters 1-inch high. If the carton wrapper used is transparent and the carton is labeled, no additional labeling will be required on the wrapper. When the product is packed for shipment to ration assembly point, only the labeled name of the product is required.

5.4.2 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

5.4.3 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

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6. NOTES

6.1 Ordering data. Acquisition documents should include the following information:

a. Title, number, and date of this document.
b. Type, styles, classes, and flavors required (see 1.2).
c. When continuous examination of formulation or procedures and controls, or both are required (see 4.3.2.1).
d. Levels of preservation and packing required (see 5.1 and 5.2).
e. When 3 pounds of types III and V product are to be packaged in a size 603 x 700 can (see 5.1.1.6).
f. Type and class of unit load required when unit loading is specified (see 5.3).
g. When type III, flavor 3 candy is to be used in Meal, Ready-to-Eat, Individual (see 3.3.2.1 and 5.1.1.4.2.3).

6.2 Use of commercial packing. Packing specified in 5.2.3 may be used for direct shipment to CONUS requisitioners for immediate use or for shipment to ration assemblers.

6.3 Supersession data. This document includes the requirements of MIL-C-10928E dated 13 February 1973 and Z-C-2104 dated June 13, 1979.

6.4 Bid sample evaluation. Bid samples will be evaluated by U.S. Army Natick Research, Development, and Engineering Center as follows. The bid sample will be rated by a consumer-type panel on a 9-point hedonic scale graduated in successive degrees of "like extremely" to "dislike extremely." For further details of the evaluation procedure, see "Hedonic Scale Method of Measuring Food Preference" from Food Technology, 1957, Vol. XI, No. 9. Copies may be obtained from the U.S. Army Natick Research, Development, and Engineering Center (ATTN: STRNC - WBF), Natick, MA 01760-5014.

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6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Preparing activity:

Army -GL

Project No. 8925-0134

Review activities:

Army - MD, TS
Navy - MC, MS
DP - SS

Civil Agencies Coordinating Activity:

USDA - MRD

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-10928F	Candy and Chocolate Confections

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

NATICK OP, 1 Apr 84

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